

Corporate Green Pledges

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We build a novel data set of time-stamped corporate decarbonization commitments—green pledges—for U.S. public firms by classifying news articles with large language models and human validation. Firms announcing green pledges tend to be larger and browner than other firms, both within and across industries. Green pledges significantly raise stock prices, consistent with a reduction in the carbon premium, and predict sizable declines in future carbon emissions and emission intensities. These effects tend to be strongest for firms in brown industries. Green pledges thus appear credible, convey relevant new information to investors, and provide meaningful financial incentives to decarbonize. (*JEL* G14, G32, Q54, Q56)

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Firms face significant transition risks as the world moves to a low-carbon economy.¹ By committing to reduce emissions, companies can

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¹ See Bolton and Kacperczyk (2021), Pastor, Stambaugh, and Taylor (2021), Ilhan, Sautner, and Vilkov (2020), and Krüger, Sautner, and Starks (2020).

mitigate their exposure to future carbon regulations and technological change. Indeed, an increasing number of corporations worldwide have announced plans and commitments to lower their carbon emissions.² Financial markets may reward such commitments if they lower perceived transition risk and the cost of capital, raising firm value. But firms may also incur substantial adjustment costs and risks, and the sign of the net effect on company values is unclear. We address this question with a new data set of time-stamped decarbonization commitments by public U.S. firms, or “corporate green pledges,” constructed from news articles using a large language model and human validation. Our event study evidence shows that green pledges significantly raise stock market valuations. Investors appear to view these pledges as credible and expect the benefits to outweigh the costs. This market response may provide some financial incentives for decarbonization. We further show that firms significantly reduce their emissions following green pledges, indicating that investors have good reasons to view these announcements as credible.

We define a corporate green pledge as a clear, new, actionable commitment to significantly reduce a firm’s future greenhouse gas (GHG) emissions. Using textual analysis, we identify the announcements of such green pledges in Dow Jones News data from 2005 to 2023, which include corporate press releases, earnings announcements, newswires and newspaper articles, and other corporate news. As green pledges are difficult to distinguish from other corporate announcements about GHG emissions, conventional text classification methods would require a large training sample of manually labeled articles and still face substantial challenges on this task. We instead classify the articles with large language models (LLMs), which enable accurate identification of green pledges without the need for training data due to their extensive pretraining, strong contextual understanding, and ability to capture linguistic nuances.³ We use GPT-4 by OpenAI—verifying robustness to the model choice using several more recent LLMs—to classify 44,605 environmental news articles. The successful use of an LLM requires careful prompt design and validation, for which we follow best practices and use human coding. In a random subsample of 1,000 articles, the classification labels from LLMs and human coders show a high level of agreement (91%). From the identified articles, and their associated ISINs and timestamps, we derive a data set of 7,166

² A 2024 report by S&P Global finds that “45% of the leading listed U.S. companies have a net-zero commitment” and that, on average, companies aim to cut Scope 1+2 emissions by 51%. The “[Net Zero Stocktake 2024](#)” reports that “nearly 60% of the 1,977 publicly listed companies we track have set net zero targets” and that the “annual revenue covered by net zero targets has increased from \$13.8 trillion in December 2020 to \$31 trillion in August 2024.”

³ Other studies have found that in empirical economic research LLMs often draw similar conclusions to humans, with the advantage of being able to quickly process large amounts of data ([Hansen and Kazinnik 2023](#); [Cook et al. 2023](#); [Beckmann et al. 2024](#); [Jha et al. 2024](#)).

firm-level green pledge events, which allows us to carry out event study analysis of the stock market effects of decarbonization commitments.

We first document new empirical facts about green pledges of listed U.S. companies. Over our sample period, about 11% of the firms made at least one green pledge, and many of those firms made multiple such announcements. The number of green pledges increases towards the end of our sample, consistent with the substantial increase in climate change concerns (Ardia et al., 2023). We validate our news-based measure by comparing it to standard measures of decarbonization commitments, such as the Carbon Disclosure Project (CDP) data. While there is significant overlap between the two data sets, our methodology identifies significantly more pledges and can track commitments as early as 2005. Green pledges are issued especially by large and brown, high-emission firms, both within and across industries. Because these types of firms are the most relevant for the green transition of the U.S. economy, their commitments to reduce emissions are particularly important.

Using event study panel regressions, we then estimate the stock market effects of our time-stamped corporate green pledges. These announcements have a positive effect on stock market valuation, which is both statistically and economically significant, with no evidence of reversal in subsequent trading days. On average, green pledges raise stock prices by 0.14–0.34%, which is a sizable increase relative to the average daily stock return of 0.015%. The estimated effects are robust to the choice of firm-level control variables, fixed effects, event subsets, sample period, and the LLMs used for classification: results based on four newer models from OpenAI, Anthropic, and Google are consistent with our baseline. We rule out the influence of confounding factors such as other corporate news released on event days, industry-specific effects, or the broader effect of environmental news on stock prices. Estimates of dynamic event study regressions show that green pledges cause immediate and persistent effects on stock market valuations, with little evidence of information leakage before or price reversals after the announcement.

Investors appear to differentiate between credible and weak green pledges. Ambitious pledges and those describing investment plans to reduce the firm's carbon footprint generate a significantly positive stock market reaction. By contrast, less ambitious pledges or those lacking specific information on how emission reductions will be achieved do not lead to significant changes in firm valuation. Furthermore, industries with high emissions—such as utilities, energy, and chemicals—have not only the highest share of pledging firms but also the strongest stock market reactions.

A straightforward explanation of our stock market results is that green pledges lower the carbon premium: As investors learn that a firm will become greener, its stock is perceived to be less exposed to climate

transition risk and/or more attractive due to nonpecuniary green investor preferences, lowering its expected return and raising its price. Green pledges could also influence stock prices through changes in expected cash flows, but we find no evidence that they improve analyst earnings forecasts or predict materially higher future profitability or dividend growth. On the whole, green pledges seem more likely to affect discount rates than expected cash flows.

Our evidence on the stock market effects is a novel contribution to the climate finance literature: positive effects from a lower carbon premium, and potentially from an improved earnings outlook, outweigh the costs and risks in the eyes of investors. While [Hartzmark and Shue \(2023\)](#) conclude that brown firms face very weak financial incentives to become more green, our results point in the opposite direction: green pledges increase stock valuations, creating positive financial incentives for companies to commit to decarbonization.

Finally, we investigate whether firms “walk the talk” by actually reducing CO₂ emissions following a green pledge, a pressing question given concerns about cheap talk and greenwashing ([Bingler et al. 2022](#)). To address it, we estimate difference-in-differences local projections for firm-level CO₂ emissions, using the methodology of [Dube et al. \(2025\)](#). The results show that green pledges are followed by statistically significant reductions in both emission levels and intensities. The reductions are quantitatively meaningful: Emissions of firms that make a green pledge are about 14% lower 5 years after the announcements, compared to firms that do not make such a commitment. Sectors with the highest carbon emissions exhibit the most substantial reductions. Since these are also the sectors with the strongest stock market reactions, the pattern suggests that pledges in carbon-intensive industries are perceived as credible and have meaningful real effects. Taken together, our findings alleviate concerns about greenwashing and indicate that green pledges tend to provide credible signals about future emissions. We do not interpret these results as evidence that green pledges causally reduce carbon emissions. Rather, pledges reveal credible private information about future emission trajectories—precisely the type of forward-looking signal that investors value and incorporate into prices.

Recent studies have investigated climate commitments using data from the CDP and the Science-Based Target Initiative (SBTi), including [Bolton and Kacperczyk \(2026\)](#), [Aldy et al. \(2023, 2025\)](#), and [Jiang \(2024\)](#). We take an alternative route and develop a news-based identification of green pledges. The resulting new database of time-stamped announcements of climate commitments allows us to use event studies to estimate the stock market response and the pricing of transition risks, in contrast to the annually reported announcements in other databases. While our news-based measure shows substantial overlap with CDP and SBTi data,

it is more comprehensive as it covers a broader range of firms and a longer sample period.⁴ Our results on the prevalence of green pledges differ somewhat from those of Bolton and Kacperczyk (2026), who find in their annual international data set that once they control for industry effects, green firms appear more likely to make commitments than brown firms. For the United States, we instead estimate that large and brown firms are more likely to make green pledges, both within and across industries, a result we confirm using both our news-based measure and the CDP commitments. Regarding the future path of emissions, our results are qualitatively consistent with those in Bolton and Kacperczyk (2026), who find that decarbonization commitments in the CDP data tend to be followed by lower emissions.⁵

In related work, Berg, Ma, and Streitz (2026) use the CDP data to show that firms often reduce emissions through divestments of pollutive assets, reallocating carbon-intensive operations rather than increasing abatement investment. Our findings provide an interesting contrast because most green pledges in our sample involve investments in low-carbon technologies, and we estimate that capital expenditures tend to rise after green pledges. This apparent contrast mainly reflects the different perspectives and methodologies: While Berg, Ma, and Streitz (2026) analyze realized changes in the emissions reported by firms and asset ownership using CDP disclosures, we focus on forward-looking corporate announcements that reveal firms' stated decarbonization intentions and associated capital reallocation.

Our paper contributes to research on climate transition risk, stock returns, and the carbon premium hypothesis of Pastor, Stambaugh, and Taylor (2021). Earlier studies have typically measured transition risk exposure using firm-level emissions and estimated the differential stock returns of green and brown firms, with mixed results. Several influential papers, including Bolton and Kacperczyk (2021, 2023), document higher brown stock returns and thus evidence for a carbon premium. Other studies instead estimate green outperformance (Garvey, Iyer, and Nash 2018; In, Park, and Monk 2019; Pastor, Stambaugh, and Taylor 2022; Bauer

⁴ CDP started recording commitments in 2013, and SBTi started only in 2015. The Net Zero Tracker (www.zerotracker.net) only captures a subset of decarbonization commitments, and does not record the announcement dates.

⁵ Various recent studies investigate other aspects of climate commitments and corporate emission targets: Acharya, Engle, and Wang (2024) develop a model of firm-level climate commitments and show evidence using the SBTi data that large firms are more likely to commit to greening. Sastry, Verner, and Marques-Ibanez (2024) study net zero commitments of banks, documenting that such commitments predict decarbonization of bank loan portfolios, but not reductions in credit supply to brown sectors or an increase in financing for renewable projects. Heeb and Kölbel (2024) show that index providers can successfully encourage corporate climate commitments. Ramadorai and Zeni (2024) document that firms' planned and actual emission abatement activities depend on prevailing beliefs about climate regulation and the political situation. Jiang, Kim, and Lu (2025) show that failures of emission targets elicit little attention based on coverage by the media.

et al. 2022). Empirical analysis of the carbon premium is typically based on past emissions data as a measure for the exposure to transition risks, but this approach has several shortcomings: Emissions data disclosure has largely been voluntary, raising the concern of selection bias; vendor estimates of emissions can be unreliable and are often revised ex post; and emissions data are available to investors only with significant lags (Aswani, Raghunandan, and Rajgopal, 2023; Zhang, 2025). While emissions data provide only backward-looking and slow-moving measures of transition risk, green pledges are forward-looking and capture new information about future firm-level greenness and transition risk exposure, which makes them better suited to studying the pricing of these characteristics and risks in financial markets. In this way, our event study results provide clear, novel evidence on the pricing of green and brown stocks.

Another key empirical challenge in climate finance is that average past stock returns tend to be poor measures of expected returns due to the short sample periods, changes in perceptions about climate risk, and mispricing.⁶ A possible remedy is to rely on estimates of expected returns for green and brown assets, as in Pastor, Stambaugh, and Taylor (2022) and Eskildsen et al. (2024). Alternatively, one can study green and brown stock returns around specific events with news about climate risk or climate policies (e.g. Engle et al. 2020; Ardia et al. 2023; Bauer, Offner, and Rudebusch 2026). We instead focus on firm-level news about future greenness, which allows us to cleanly identify the impact on stock market valuations and draw conclusions about the carbon premium and the pricing of transition risks.

Many studies have used textual analysis to measure climate risks, typically by constructing aggregate indices from newspaper articles. For instance, Engle et al. (2020) use news articles from *The Wall Street Journal* to build a climate news index, Ardia et al. (2023) construct a news-based index of climate change concerns, and Faccini, Matin, and Skiadopoulos (2023) derive several different climate risk measures from news articles. Only a few studies have used text methods to investigate the pricing of climate risk at the firm level. Sautner et al. (2023a) and Li et al. (2024) use companies' earnings calls to measure firm-level climate change exposure, and Sautner et al. (2023b) document changes in the risk premium associated with such exposure. Dzieliński et al. (2023) investigate the response of stock prices and future GHG emissions to discussions of climate-related topics in earnings calls, and find reductions in future emissions as evidence, consistent with our results, that firms "walk the climate talk." Our measure differs from earlier ones in that it captures

⁶ Atilgan et al. (2023) incorporate earnings announcements in their analysis and conclude that the carbon premium in their data sample in fact reflects unexpected returns and mispricing.

news about future firm-level emissions and thus changes in greenness and transition risk.

In the climate finance literature, there is an active debate about greenwashing in corporate announcements, and, more specifically, cheap talk in climate commitments (Nemes et al. 2022; Bingler et al. 2022; Bingler et al. 2024; Dzielinski et al. 2023; Sastry, Verner, and Marques-Ibanez 2024). Our evidence that green pledges predict reductions in future emissions and emission intensities contributes to this debate. It suggests that corporate green pledges are at least partly credible and should not be discarded wholesale as cheap talk.

Our work is closely related to other studies in climate finance and the environmental, social, and governance (ESG) literature using event studies to estimate the effects of firm-level news on stock prices and firm prospects. Krüger (2015) studies how stock prices react to positive and negative news regarding a firm's corporate social responsibility (CSR), based on an identification of CSR events from text data. Derrien et al. (2025) document that negative ESG news, measured by RepRisk, tends to lower analyst forecasts for earnings and sales, which is evidence for a cash flow channel in this context. We contribute to this literature by documenting the stock market effects of firm-level news about CO₂ emissions and providing evidence for the importance of a discount rate channel and changes in the carbon premium.

1. Identification of Green Pledges

Our starting point is to define and identify decarbonization commitments of U.S. firms using text data. We use Dow Jones News, a large data set of newswire and newspaper articles from Dow Jones, *The Wall Street Journal*, Barron's, MarketWatch, and other sources. These news outlets, and especially the newswires, are closely monitored by financial market participants. The data also include press releases, earnings announcements, and other corporate news. Versions of this text data set have been used in a number of studies in finance and economics.⁷ The text data comes with a variety of attributes, including the geographical regions covered by the news, categories of subjects, and the precise time-stamps. Additionally, Dow Jones assigns a list of ISINs to each article, which simplifies the process of linking them to firms. Our sample period ranges from January 2005 to December 2023. We refine our selection of articles by including

⁷ Ke, Kelly, and Xiu (2020) construct a novel sentiment score from this text data, which they use to predict stock returns. Barbaglia, Consoli, and Manzan (2023) used data from Dow Jones to analyze the gains of using sentiment measures in macroeconomic forecasting. Ravenpack uses Dow Jones Newswires data to construct its popular sentiment scores, which have been used widely in finance and economics (e.g. Kim 2022; Audrino and Offner 2024). See Schmanski, Scotti, Vega, and Benamar (2025) for text analysis using this data in the context of monetary policy communication.

only those related to companies in the United States and categorized as environmental news.⁸ The resulting sample includes 44,605 news articles and announcements.

To accurately classify green pledges, we first need a workable definition. We define a green pledge as an announcement of a new, clear, and actionable commitment by a firm to significantly reduce future direct greenhouse gas emissions. This definition aims to identify articles with new information about a firm's expected future emissions, which is relevant for stock market investors. Updates to prior commitments are included only when they introduce more stringent targets or new initiatives. For example, a firm advancing its carbon neutrality target by reallocating capital to renewable energy would qualify as a green pledge. By contrast, statements that merely reaffirm existing goals do not. Green pledges should be about direct emissions because these are easier to quantify and directly attributable to firm-level decisions, making them the most reliable indicator of a company's decarbonization efforts. We focus on official corporate communications—such as press releases, regulatory filings, and other corporate announcements—and news coverage of a company's climate commitments. Overall, the announcement should contain material new information for financial market participants and the broader public about expected firm-level emissions.

Our goal is to accurately classify the news articles into those that contain corporate green pledges and those that do not. A wide range of text classification methods could, in principle, be used for this classification problem.⁹ However, most methods would struggle with this task because green pledges sound superficially similar to other GHG-related corporate announcements. Achieving adequate accuracy would require a large volume of labeled training data—many thousands of labeled articles—and careful model tuning.

We instead use large language models to classify the news articles. By now, LLMs have been successfully used in a wide range of applications in empirical economic research.¹⁰ Our application requires a suitably chosen prompt with accurate and detailed instructions that are consistent with our definition of a green pledge. In addition, it is necessary to carefully evaluate the results, given the possibility of hallucination, bias, restrictions

⁸ Specifically, we use articles tagged with at least one U.S. ISIN, the U.S. indicated as the geographical region, and “environment” as the subject classification, using the metadata provided by Dow Jones.

⁹ For excellent treatments of text analysis methods and their applications in economics and other social sciences, see [Grimmer, Roberts, and Stewart \(2022\)](#), and [Ash and Hansen \(2023\)](#).

¹⁰ [Lopez-Lira and Tang \(2024\)](#) find that GPT-4 can make accurate stock market predictions based on news headlines. [Hansen and Kazinnik \(2023\)](#) use GPT models to classify the monetary policy stance based on language in FOMC statements, and find logical reasoning similar to human coders. Large language models have been used to evaluate the information in company earnings calls by [Cook et al. \(2023\)](#) and [Beckmann et al. \(2024\)](#), and to evaluate corporate policies by [Jha et al. \(2024\)](#). See [Korinek \(2023\)](#) and [Ash and Hansen \(2023\)](#) for other applications of LLMs in economics.

from content use policies, and other related problems of LLMs. For both our prompt design and for evaluation of the LLM classifications, we make extensive use of human coding. Our approach is similar to that of [Eloundou et al. \(2024\)](#), who also use human labeling and LLMs, fine-tuning a prompt to yield good agreement between both. Our main results are based on OpenAI's GPT-4, but we also obtain classifications from several more recent models to verify robustness.

The first step of our text classification was to design a suitable prompt using an iterative process. We started with a simple prompt based on our definition of a green pledge, and then refined the prompt in successive iterations, each time using the model to classify a subset of articles that were also given to human coders and comparing human- and model-based classifications. We also asked the LLM for a concise justification for each classification decision, which offers insights into the model's reasoning process and its comprehension of the assignment. Based on these results, we then iteratively modified the prompt, typically making the criteria more stringent to avoid false positives. The result of this process was the following final prompt:

Classify the following article as positive or negative depending on whether it contains an announcement that the company will reduce its future emissions of greenhouse gases, such as carbon dioxide. Classify an article as positive only if the company announces a significant reduction of direct emissions, that is, emissions that occur from sources controlled or owned by the company. The announcement should be news and should describe the company's commitments and plans for the future. Do not classify articles as positive that only contain announcements to reduce indirect emissions, that is, emissions that a company causes indirectly from the energy it purchases and uses. Also do not classify articles as positive if they are only about past performance, about a corporate social responsibility (CSR) report describing past emission reductions, about other environmental measures such as waste reduction, use of recycled paper, or planting trees, or announcements by the government. If an article is empty, or does not contain enough information, classify it as negative. Answer "YES" for positive articles and "NO" otherwise.

The second step was to classify our entire corpus of news articles. The specific GPT-4 model we used for our baseline results was gpt-4-0613, the latest GPT-4 variant available at the time of our analysis. In order to have a model-based classification that is deterministic, given the text input, we set

the temperature, a key parameter of any LLM, to zero.¹¹ We used OpenAI's Python API to classify the entire data set of 44,605 news articles.

In the following, we present three examples of green pledges identified by GPT-4. They demonstrate the successful identification of announcements that contain commitments to reduce GHG emissions:

- October 14, 2009: Wells Fargo & Company (NYSE: WFC) announced today that it has set a goal to reduce its U.S.-based greenhouse gas emissions by 20 percent below 2008 levels by 2018. The Company is focusing on reducing its carbon footprint as part of its continued environmental commitment to lead by example and to fulfill its pledge as a member of the U.S. Environmental Protection Agency's (EPA's) Climate Leaders program, which Wells Fargo joined last year. [...]
- January 16, 2020: Microsoft Corp. on Thursday announced an ambitious goal and a new plan to reduce and ultimately remove its carbon footprint. By 2030 Microsoft will be carbon negative, and by 2050 Microsoft will remove from the environment all the carbon the company has emitted either directly or by electrical consumption since it was founded in 1975. [...]
- July 21, 2020: Apple today unveiled its plan to become carbon neutral across its entire business, manufacturing supply chain, and product life cycle by 2030. The company is already carbon neutral today for its global corporate operations, and this new commitment means that by 2030, every Apple device sold will have net zero climate impact. [...]

The third and final step was to validate the model-based results using human coders. This step is crucial in applied economic research, as it provides an essential benchmark for the accuracy of LLM results (Dell 2025; Ludwig, Mullainathan, and Rambachan 2025). We selected a random subset of 1,000 articles and tasked two research associates, who were not otherwise affiliated with this project, each to label half of this validation sample. As instructions to our human coders, we simply provided them with the same prompt we gave to the LLM.

Table 1 shows a confusion matrix comparing the labels of GPT-4 with those of the human coders. We calculate commonly used statistics for binary classifications, treating the human labels as the "truth" for this purpose. The accuracy of the GPT labels, defined as the share of identical classifications between GPT and human coders, is 91%. This high level of

¹¹ See Beckmann et al. (2024) for more details on this issue. Note that a zero temperature parameter does not guarantee reproducibility of the results because the model is not open source and OpenAI could well modify or further fine-tune it.

Table 1
Comparison of classification by GPT-4 and human coders

		Human Coder		Total
		Negative	Positive	
GPT	Negative	838	19	857
	Positive	67	76	143
Total		905	95	1,000

Confusion matrix for the classifications of 1,000 randomly selected news articles by the model GPT-4 and human coders.

agreement partly reflects the large number of negative cases. The precision, defined as the fraction of GPT-positive articles that were also classified as positive by human coders, is 53%, reflecting that human coders applied stricter criteria: they identified 95 news articles as green pledges, while GPT classified 143 as positive. The sensitivity (or recall) is 80%, implying that of the 95 pledges identified by human coders, 76 were also recognized as positives by GPT. Hence, GPT successfully identifies a large fraction of the pledges detected by human coders. Overall, the human validation suggests that GPT-4 classifies most green pledges accurately, with few false negatives. However, its classifications are somewhat less conservative than those of the human coders, implying the presence of some false positives.

The landscape of LLMs is quickly evolving, and OpenAI has released many new models since we initially carried out the classification with GPT-4 and wrote the first version of this paper. To assess whether newer models yield similar results, we have classified the validation sample of 1,000 articles with several recent models from OpenAI, Anthropic, and Google. The results, which are reported in [Internet Appendix A.1](#), show that different newer models tend to classify somewhat fewer articles as green pledges, similar to the tendency of human coders to be more selective, and that there is some notable degree of disagreement across models.¹² A comparison to the human labels shows that newer models tend to have higher precision but lower recall, and that GPT-4 remains overall quite competitive for green pledge classification. Importantly, we will consider labels from different LLMs to assess the robustness of our main stock market event study results in [Section 3](#).

Our validation exercises confirm that identifying green pledges from news articles is challenging. Standard econometric theory can speak to the likely consequences of imprecise measurement of green pledges for our downstream analysis. The event indicator variable that identifies the time period and firm of a green pledge necessarily contains some measurement error. This variable is a regressor in our event study regressions for stock returns in [Section 3](#) and in the difference-in-difference estimation

¹² In additional analysis, we find that disagreement is especially high for longer and more recent articles, as well as for articles on the subject of carbon emissions.

Table 2
Categories and ambition scores of green pledges

		Ambition		Total
		Limited	Significant	
Category	Investment	262	5,799	6,061
	Divestment	10	90	100
	No specifics	669	32	701
Total		941	5,921	6,862

Distribution of green pledge articles according to category label and ambition score.

in [Section 4](#). Under the assumption that this measurement error is uncorrelated with both the true underlying green pledge indicator and the regression error term, the setting is analogous to a classical errors-in-variables problem.¹³ In that case, the resulting bias would attenuate our downstream estimates towards zero, providing a conservative estimate of the true effect of green pledges on stock returns and emissions. More generally, the robustness of our main event study results across multiple LLMs alleviates concerns about measurement error.

The end result of our classification with GPT is a selection of 6,862 news articles—about 15% of all environmental articles on U.S. public companies—that are deemed by GPT to contain green pledges. Before turning to the analysis of the characteristics and effects of green pledges in the rest of the paper, we can go further with this text data and use LLMs to try to understand the nature and ambition of these commitments.

Firms can reduce their GHG emissions in a number of different ways, but broadly they have the choice of investing in new low-carbon technologies or divesting from carbon-intensive assets. We use an LLM to categorize the articles that were identified as green pledges, according to whether the announcement contains (1) information about new investments to reduce emissions, such as green energy technologies; (2) information about divestments from pollutive assets, such as the sale of high-emission business units; or (3) no specific information on how emission reductions will be achieved.¹⁴ [Table 2](#) shows that the vast majority (88%) of green pledge articles indicate that firms would invest in new technologies to reduce emissions.

Another important question about decarbonization commitments is whether they are actually meaningful and ambitious, or whether they announce only minor reductions in order to avoid missing the targets. We try to address it by having the model score the level of ambition for the emission reduction as either (1) limited or vague commitment

¹³ While the mean of the measurement error is positive if we have more false positives than false negatives, this would only bias the intercept of the regressions.

¹⁴ The detailed prompt is given in [Internet Appendix A.2](#). We used GPT-4o for this task.

with minor expected impact on emissions or (2) significant and specific commitment with substantial impact. As shown in Table 2, most articles were judged to contain announcements of significant and specific plans for emission reductions. Judging from the announced green pledges, the level of ambition would appear to be quite high.

For the following analysis, we identify company-level green pledge events from the identified articles. Each of these articles can tag multiple firms (ISINs), for example, if certain decarbonization initiatives involve several companies. In addition, on a given day there may be multiple identified articles for the same firm, for example, if a press release is followed by coverage in a news article. For our empirical analysis, we create unique firm-date events, based on combinations of company ISINs and article date, to represent corporate green pledges. The result is a sample of 7,166 green pledge events.

For many firms, our sample contains more than one green pledge. A simple approach to reduce the noise in our classification is to only consider the first green pledge for each of the 1,528 firms that have made at least one pledge. For the stock market analysis in Section 3, we will consider both types of green-pledge indicators, either using all events for a firm or using only the first identified green pledge for each firm.¹⁵ For estimating the relationship with future emissions in Section 4, we use a difference-in-differences approach with an indicator variable that turns on when a firm makes its first pledge.

2. Green Pledges over Time and across Firms

In this section, we document patterns of corporate green pledges over time and across firms. In addition, we compare and validate our text-based pledges against other databases, such as the data from the Carbon Disclosure Project.

Figure 1 plots the annual number of (environmental) news articles in our text data, and the number of articles that the LLM identified as containing green pledges. Our data contained a significant number of environmental news articles each year, with a peak in 2010 and a strong increase in recent years. The high number of environmental articles around 2010 can be

¹⁵ In addition, we also consider two other subsamples of green pledges that impose a more stringent event definition to address two other concerns: First, duplicate or follow-up articles can appear in subsequent days. To address this issue, we require a certain number of days to elapse between two successive events for the same firm. Choosing the required distance as too wide could result in excluding new commitments, while setting it too narrowly leads to duplicate events concerning the same green pledge. As a middle ground, we choose a 30-day distance, which we find a reasonable distance to avoid duplicates while not losing new commitments. The second issue arises as articles are sometimes tagging multiple companies. To avoid matching articles to firms that were mentioned in the articles but were not their primary subject, we follow Ke, Kelly, and Xiu (2020) and include only articles tagged with a single company in our analysis. Our “30-day distance” sample contains 5,898 events, and our “single-tag” sample contains 5,237 events.

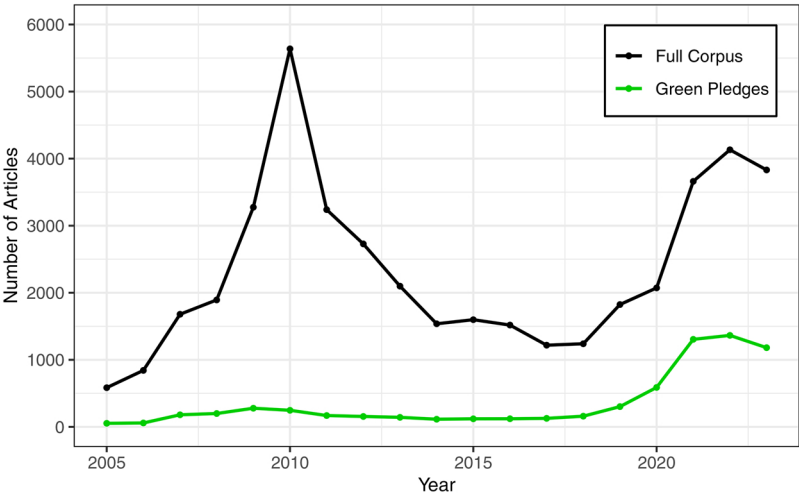


Figure 1
Number of environmental news and green pledge articles over time
Yearly number of articles in the full text corpus of environmental news articles (black line) and classified as green pledge by GPT (green line). The total number of articles is 44,605 articles, of which 6,862 are classified as green pledges. Sample period: 2005 to 2023.

attributed to the 2009 Copenhagen UN Climate Change Conference and the explosion of Deepwater Horizon in April 2010.¹⁶ Over the early part of our sample period, between 2005 and 2018, the number of green pledge articles was around 150 per year. Towards the end of the sample, around 2019, the number of identified articles started increasing substantially, and each year since 2021 there were over 1,000 green pledge articles. This pattern may be due to increased public attention to the risks of climate change—as evidenced for example in the text-based measure of climate change concerns of [Ardia et al. \(2023\)](#)—and increasing climate transition risks that cause more companies to commit to climate mitigation and emission reductions. By contrast, during the years around the Paris Agreement in 2015 there was no appreciable increase in green pledges, suggesting that this international initiative actually had only a modest effect on the corporate sector.

Turning now to the firm-level announcements, a crucial question is how our measure of green pledges compares to those recorded in databases of corporate climate commitments. To address it, we carry out a comparison with the CDP data that has been widely used in research in climate finance

¹⁶ When computing term-frequency inverse document-frequency (tf-idf) for the year 2010, the most frequent terms are “bp” and “oil.” In contrast, the most frequent terms across the entire sample period include “statements,” “water,” and “energy.” This disparity underscores the significant impact of the Deepwater Horizon spill on news coverage in 2010.

and climate economics.¹⁷ This database, which begins in 2013, tracks firms' emission reduction targets on an annual basis. Since emission targets are continually updated in the CDP data, we focus on the first recorded CDP commitment for each U.S. firm, which we can compare to the first green pledge identified from our text data. There are 809 U.S. firms in our CDP data set, of which 547 also have a green pledge in our text data, meaning that we have a substantial overlap of 68%.¹⁸ Our method detects about twice as many firms with green pledges, suggesting that our approach delivers a more comprehensive record of decarbonization commitments than the CDP database. **Figure 2** plots annual series of the cumulative number of first decarbonization commitments in both data sets. We identify pledges as far back as 2005, while CDP data start only in 2013, so we can significantly extend the available sample period. Between 2013 and 2020, both data sets show a broadly similar increase in first green pledges.¹⁹ Since 2020, however, we observe a sharp rise in pledges captured by our text-based approach, likely reflecting heightened climate awareness and increased corporate climate action, which the CDP data does not pick up to the same extent. Further analysis of the green pledges that are recorded in both data sources shows that there is a close correspondence of firm-level climate commitments in our text data and in the CDP data, and, if anything, that our text-based method identifies these commitments in a more timely fashion.²⁰ Overall, there are similar trends and significant overlap between the two data sources, but our text-based method provides broader coverage and a longer sample period. Of course, another important advantage of our text-based approach is that it includes an exact timestamp for each announcement, which is essential for estimating the stock market effects of climate commitments.

In the following, we analyze the distribution of green pledges across industries and the role of firm characteristics. We focus on the question of what kind of firms are most likely to commit to decarbonizing their business models. For the transition to a low-carbon economy, it is particularly important that large, high-emission firms commit to reducing their carbon footprint. If, instead, green pledges are announced predominantly by firms that use green technology already, or by firms that

¹⁷ The Carbon Disclosure Project advertises itself as “the world’s largest primary corporate environmental data set.” See <https://www.cdp.net/en/data> for more information. Studies using this data include, among many others, Bolton and Kacperczyk (2021, 2023, 2026) and Ramadorai and Zeni (2024).

¹⁸ A similar comparison with the SBTi database reveals that among the 400 U.S. firms in that data set, 305 (76%) were identified as making a green pledge in our text data. That is, the overlap of our green pledges with SBTi commitments is even larger than with CDP commitments.

¹⁹ The similar upward trend shown in **Figure 2** also alleviates concerns that the higher number of green pledges in our text-based data is due to false positives.

²⁰ **Internet Appendix Figure B.2** shows that these green pledges are most likely to occur either at a similar time in both data sets, or earlier in the text data than in the CDP data.

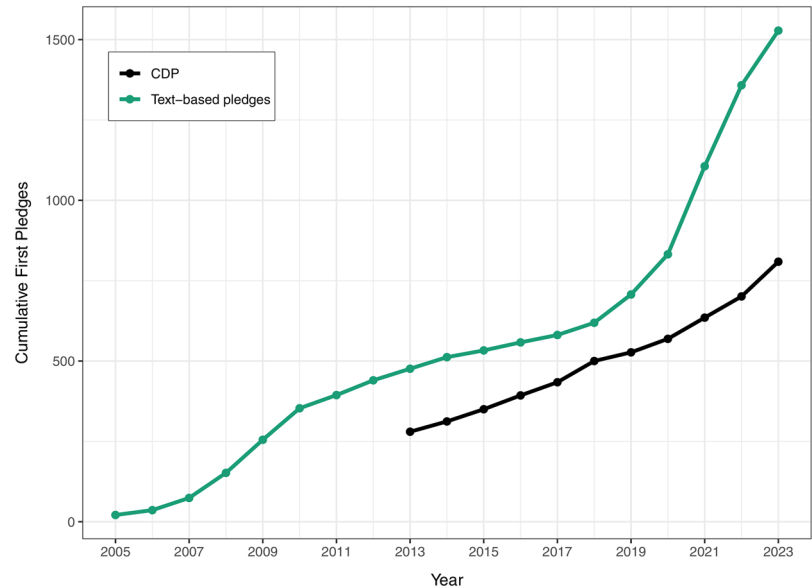


Figure 2
Cumulative first pledges: Text-based vs. CDP targets
The figure plots the cumulative number of firms’ first green pledges identified with our text-based approach alongside the cumulative number of first emission reduction targets reported in the CDP database. Sample: 2013 to 2023.

have low emissions in the first place and hence a low exposure to transition risks, the aggregate impact on emissions would be small. In this case, green pledges would be less useful for the transition of the U.S. economy to a low-carbon future.

As the following analysis requires firm characteristics, we merge the data on green pledges with data from financial reports and emissions (both of which are annual). Our accounting data come from Compustat and our emissions data are from S&P/Trucost. We use two emission variables: level of emissions, defined as the sum of Scope 1 (direct) and Scope 2 (indirect) emissions (in million tons of CO₂ equivalents), and emission intensity defined as the sum of Scope 1 and Scope 2 emissions divided by revenue (in million USD). Following earlier work in climate finance, we exclude Scope 3 emissions because these indirect emissions from upstream and downstream activities of the reporting firm are very large in magnitude and particularly difficult to estimate (Bauer et al. 2022; Huij et al. 2024). Internet Appendix B provides detailed descriptions and summary statistics for the variables based on accounting financials and emissions.

After the data merge, our sample contains 5,603 green pledge events, corresponding to 995 different firms. This number of firms with climate commitments corresponds to 11% of the 8,882 public U.S. firms in our

Table 3
Pledges and emissions across industries

Industry	Firms	Pledges/firm	Share pledged	Emissions	Intensity
Utilities	145	8.03	0.66	20.63	2.38
Energy	337	1.27	0.22	8.01	0.61
Chemicals	167	1.92	0.29	2.87	0.48
Other	1,156	0.81	0.12	2.43	0.23
Manufacturing	631	1.05	0.21	1.49	0.22
Nondurables	342	0.85	0.20	0.83	0.10
Retail	693	0.55	0.12	0.81	0.07
Telecommunication	230	0.34	0.05	0.68	0.02
Durables	190	1.67	0.14	0.59	0.04
Business Equipment	1,733	0.37	0.11	0.37	0.04
Health Care	1,635	0.06	0.03	0.14	0.04
Finance	1,611	0.17	0.05	0.06	0.01

Summary statistics for firms, pledges, and emissions across 12 Fama-French industries. For each industry, the table shows the total number of firms, the number of pledges per firm, the share of firms with pledges, the average level of carbon emissions (Scope 1+2, measured in million tons of CO₂), and the average emission intensity (in kilotons of CO₂ per USD million of revenue). Sample period: January 2005 to December 2023.

Compustat sample.²¹ The number of green pledges varies significantly across firms. It is most common for firms to make only one pledge, and this is the case for 392 firms. But the median number of pledges per firm is two, and the distribution has a long right tail.²² The fact that some firms made several pledges, sometimes in double digits, is an important reason why we also consider restricted subsamples of green pledge events in our subsequent analysis in order to avoid duplicates and follow-up articles, such as the sample of first pledges.

To study the cross-industry distribution of firms and green pledges, we use the 12 Fama-French industries, which are based on a mapping of Standard Industrial Classification codes into distinct economic sectors and are widely used in financial market research.²³ Table 3 presents industry-level summary statistics with industries ordered by the emissions from brown to green. We report the number of firms and green pledges per industry, the number of pledges per firm and share of firms with pledges, as well as average firm-level emissions and average emission intensities.

²¹ We ensure that the set of firms is identical to the one used in Section 3, where we also merge with CRSP stock market data. From the original 1,528 firms with green pledges, we retained 995 in our final sample. This reduction arises for two reasons: first, CRSP does not cover all securities, such as OTC stocks; second, we filter out securities that are not ordinary or Class A common shares, as well as those not traded on the NYSE, AMEX, or NASDAQ.

²² Among the firms that made at least one green pledge, the median is 2, the mean is 5.6, and the 95th percentile is 22 green pledges per firm. See also Internet Appendix Figure B.1 for the distribution of the number of pledges across firms.

²³ For the purpose of controlling for industry effects in our empirical analysis—using fixed effects, for example—we will rely on the more granular 48-industry Fama-French classification. The two classifications are consistent with each other in a hierarchical sense: the 12-industry classification aggregates the 48 industries into broader economic sectors.

Table 4
Characteristics of firms with and without green pledges

	Pledge	No Pledge	<i>p</i> -value	<i>t</i> -statistic
Emissions	3.85	0.16	0.00	7.16
Emission Intensity	0.43	0.08	0.00	7.44
Size	22.94	1.49	0.00	10.51
Book-to-market	0.50	0.69	0.00	−10.85
Leverage	5.69	6.46	0.27	−1.11
Sales growth	0.49	1.87	0.05	−1.95
Return on equity	0.17	−0.66	0.00	3.77

Summary statistics for firms with and without green pledges. For both groups, sample averages of the different firm characteristics are reported as well as *p*-values and *t*-statistics for the differences in means between the two groups. Firm characteristics are winsorized at the 1%/99% level.

The utilities sector stands out as being both the brownest sector—with the highest firm-level emissions and emission intensities—and having by far the highest number of green pledges, averaging 8.03 pledges per firm. Other carbon-intensive industries, such as energy, chemicals, and manufacturing, also record relatively high levels of pledging activity, with 1.27, 1.92, and 1.05 pledges per firm, respectively. In contrast, firms in low-emission industries such as finance, health care, and business equipment show far fewer pledges, averaging only 0.17, 0.06, and 0.37 pledges per firm, respectively. This tendency for brown industries to have more frequent pledges is also evident in the share of firms that have made any green pledges in our sample. For instance, in the utilities sector, 66% of firms have announced a green pledge, compared to 5% in the finance sector. Overall, these results indicate that firms in high-emission, high-intensity sectors are significantly more likely to announce decarbonization commitments than those in greener industries.

To study the firm-level characteristics associated with green pledge announcements, Table 4 compares the attributes of firms with and without such pledges. Specifically, the table reports for both groups the average firm characteristics, first averaged over time for each firm and then across firms within each group. The final two columns display the *p*-values and *t*-statistics testing for differences in the cross-sectional means of these characteristics. All variables are winsorized at the 1%/99% level to reduce the influence of outliers.

In line with the industry-level analysis, Table 4 shows that firms with a green pledge have, on average, significantly higher emissions and higher emission intensities. In addition, these firms tend to be larger. The differences are both economically and statistically significant. This finding alleviates the concern that decarbonization commitments are primarily made by green firms, with a resulting smaller effect on aggregate emissions. We also find significant differences for book-to-market value, sales growth, and return on equity: Firms with green pledges tend to have higher valuations, lower sales growth, and higher profitability.

Bolton and Kacperczyk (2026) raise the concern that while unconditionally brown firms appear more likely to make decarbonization commitments, within industries greener firms appear to be more likely to make such commitments, according to their analysis. The authors conclude that it is the greener firms, which have likely already adapted their business models to reduce emissions, that make green pledges, and that, consequently, such pledges may have only modest effects on aggregate emissions and a small role in the green transition.

To address this issue, and to get a more nuanced picture of the type of firm that tends to make green pledges, we estimate panel regressions that control for industry effects and firm characteristics, using a firm-by-year panel data set. The dependent variable is a binary indicator that takes the value of one in the year a firm makes its first green pledge and remains one in the subsequent years, following the definition of Bolton and Kacperczyk (2026). The regressions include lagged firm characteristics (from the previous year) and year fixed effects. We consider specifications without and with industry fixed effects based on the Fama-French 48-industry classifications.

The results in Table 5 clearly demonstrate that brown firms and large firms are more likely to make green pledges, even after controlling for industry fixed effects and various firm-level characteristics. The effects of log emissions and size on the issuance of a green pledge are positive and statistically significant at the 1% level for all specifications. These results are robust to the use of only reported emissions data, excluding vendor estimates of firm-level emissions; in fact, the findings become even stronger when considering disclosed emissions only.²⁴

Our findings in Table 5 appear to be at odds with the evidence presented by Bolton and Kacperczyk (2026), who find that in panel regressions with industry fixed effects green firms are more likely to make decarbonization commitments. There are two fundamental differences between the data set used in Bolton and Kacperczyk (2026) and this paper. First, they identify green pledges by using firms that sign up for carbon initiatives such as the CDP and the SBTi, while we use green pledges identified from newspaper articles. Second, Bolton and Kacperczyk use an international sample of firms while we focus on the U.S. stock market. To better understand the source of the discrepancy, we revisited the analysis of Bolton and Kacperczyk (2026) using CDP commitments for a sample restricted to U.S. firms. These estimates using CDP pledges of U.S. firms closely mirror our

²⁴ Separate results using only disclosed emissions or only Trucost's estimates of firm-level emissions are reported in Internet Appendix Table B.3. Furthermore, we estimated regressions using the same set of controls used in Bolton and Kacperczyk (2026), and found that this leads to the same conclusions as the estimates in Table 5. As some of these controls are not available for all firms, we only use the set of controls reported in the table for our main analysis. Our findings also remain essentially unchanged when using GICS six-digit industry classification for the industry fixed effects, as in Bolton and Kacperczyk (2026).

Table 5
Green pledges and within-industry variation of firm characteristics

Model	(1)	(2)	(3)	(4)	(5)	(6)
Log Emissions	0.06*** (0.004)	0.06*** (0.003)			0.04*** (0.002)	0.03*** (0.002)
Size			0.04*** (0.01)	0.03*** (0.01)	0.06*** (0.001)	0.07*** (0.002)
Book-to-market					4.02** (2.05)	4.82* (2.54)
Leverage					0.15*** (0.04)	0.12*** (0.04)
Return on equity					−0.01*** (0.001)	−0.01*** (0.001)
Sales growth					0.003 (0.01)	0.01 (0.01)
Year FE	Y	Y	Y	Y	Y	Y
Industry FE	N	Y	N	Y	N	Y
Observations	20,915	20,915	66,452	66,452	18,930	18,930
R ²	0.21	0.26	0.16	0.21	0.26	0.30

Panel regressions for a yearly green pledge indicator variable, which is equal to one starting in the year of a firm's first green pledge and thereafter. Independent variables are firm characteristics lagged by 1 year. Controls are book-to-market, leverage, return on equity, and sales growth. Columns (2), (4), and (6) include 48-industry Fama-French fixed effects. Standard errors, which are clustered by firm and year in each model, are given in parentheses. Values are winsorized at the 1%/99% level. * $p < .1$; ** $p < .05$; *** $p < .01$. Sample period: January 2005 to December 2023.

findings for text-based green pledges in [Table 5](#).²⁵ Thus, the differences between our results and those in [Bolton and Kacperczyk \(2026\)](#) may be due to their use of a global sample of stocks.

This section has provided new evidence on the distribution of green pledges over time, across industries, and across firm characteristics. Green pledges have occurred in significant numbers since the beginning of our sample in 2005, they have increased over time, and become particularly widespread since around 2020. We observe significantly more green pledges in high-emission sectors. Both within and across industries, brown and large firms are more likely to make green pledges.

3. Stock Market Effects

The key question of our paper is how stock market valuations react to corporate green pledges. This is an empirical question because the sign of this reaction is a priori not clear. Positive effects on valuations can result from a lower trajectory of projected emissions that reduce the carbon premium, due to lower transition risk exposure or a stronger appeal to green investors ([Pastor, Stambaugh, and Taylor 2021](#)). However, the existence and size of the carbon premium have not been firmly established ([Bauer et al. 2022](#); [Zhang 2025](#)). The effects of decarbonization on future profits and dividends could go either way: If green pledges signal that firms

²⁵ See [Internet Appendix Table B.2](#) for these estimates.

managed to get access to profitable green investment projects, this could cause positive effects. But decarbonization projects may require significant upfront investments and potentially far-reaching changes to a company's production processes and business model. The transition to lower or net-zero emissions could be costly, weighing on the outlook for earnings and dividends. In addition, the risks involved in this transition could actually raise the required risk compensation and cost of capital. There is no clear theoretical prediction for the net effects of green pledges on either expected dividends or discount rates.

We use event studies to identify the effects of the announcements of corporate green pledges on stock prices. The identification assumption is that the corporate announcement is predetermined with regard to the company's stock return on the day of the announcement. By making this assumption, we follow a long tradition in empirical asset pricing of using event studies to identify the stock market effects of corporate news.²⁶ The announcements of green pledges provide new information to investors about a company's trajectory and strategy for GHG emissions, and we estimate the stock price response to this new information.

Our analysis uses daily stock market data for U.S. firms from CRSP, including all common equity listed on NYSE, AMEX, or NASDAQ. To accurately assign a corporate green pledge to a trading day, we take into account the timestamp of the news article; announcements made after NYSE closes (at 4:00 p.m. Eastern time) are assigned to the subsequent trading day. We use several standard control variables in our event study, including size, book-to-market ratio, leverage, sales growth, and return on equity. For those control variables that require annual accounting data, we use a publishing lag of 4 months, following common practice in empirical asset pricing, to ensure that the information was available to investors at the time of the green pledge.²⁷ [Internet Appendix B](#) contains a detailed description and summary statistics of the control variables.

We estimate the panel regression

$$R_{it} = \beta d_{it} + \gamma X_{it} + \alpha_s + \delta_t + \epsilon_{it}, \quad (1)$$

where R_{it} is the stock return of company i on day t , d_{it} is an event dummy that equals one if the firm announced a green pledge on this day, X_{it} includes the firm-specific control variables, α_s are industry fixed effects based on the 48 Fama-French industries, δ_t are day fixed effects, and ϵ_{it} is the residual. In alternative specifications, we use firm fixed effects instead of industry fixed effects to assess the robustness of our results. Standard errors are clustered by firm and day in all our panel regressions.

²⁶ For surveys of this event study literature within empirical asset pricing, see [MacKinlay \(1997\)](#) and [Kothari and Warner \(2007\)](#).

²⁷ For example, the accounting data for 2022 is used for stock returns from May 2023 onwards.

Table 6
Stock market response to green pledges

	All green pledges			First green pledges		
	(1)	(2)	(3)	(4)	(5)	(6)
Green pledge	0.135*** (0.037)	0.150*** (0.038)	0.205*** (0.040)	0.326*** (0.104)	0.335*** (0.104)	0.338*** (0.100)
Size	0.007*** (0.002)	−0.099*** (0.008)		0.007*** (0.002)	−0.099*** (0.008)	
Book-to-market	−7.968* (4.273)	−23.177*** (5.307)		−7.957* (4.273)	−23.179*** (5.307)	
Leverage	−0.198 (0.184)	−0.520 (0.376)		−0.197 (0.184)	−0.520 (0.376)	
Sales growth	−8.190*** (3.063)	0.652 (2.879)		−8.194*** (3.064)	0.653 (2.879)	
Return on equity	44.760*** (6.471)	15.262*** (4.826)		44.751*** (6.470)	15.258*** (4.826)	
Number of pledges	5,012	5,012	5,603	842	842	990
Observations	14,788,589	14,788,589	17,511,046	14,788,589	14,788,589	17,511,046
R ²	0.189	0.190	0.176	0.189	0.190	0.176
Industry FE	Y	N	Y	Y	N	Y
Firm FE	N	Y	N	N	Y	N
Day FE	Y	Y	Y	Y	Y	Y

Panel regressions for daily stock returns. Columns (1)–(3) show results for an event dummy that equals one on all days that a firm announces a green pledge, and columns (4)–(6) show results for a dummy that equals one only on the day of the first green pledge of each firm. Alternative specifications use fixed effects (FE) for industry, based on 48-industry Fama-French classification, firm, and trading day. Standard errors, clustered by firm and trading day, are given in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$. Sample period: January 2005 to December 2023.

Table 6 shows the estimation results. The first three columns report the estimates for regressions using the full set of green pledge events, using different fixed effects and firm-level controls. Across all three specifications, the estimated effect is positive and statistically significant at the 1% level. The size of the estimated coefficient indicates that the daily stock return increases by between 0.14 and 0.21 percentage points when a firm announces a green pledge. The average daily return in our sample is about 0.015% (see Table B.1), meaning that on event days returns are about 10 times larger than on nonevent days. In other words, the estimated effect is both statistically and economically highly significant. Furthermore, if there is attenuation bias in estimates of β as a result of the measurement error in d_{it} discussed in Section 1, the true stock price impact of corporate green pledges may be even larger than these estimates.

The last three columns of Table 6 report the estimates for regressions using only the first green pledge for each firm. In these regressions the estimate for β is in the range of 0.33 to 0.34 percentage points, substantially larger than for the regressions using all pledges. Clearly, the estimated positive effect is particularly strong for the first green pledge issued by a firm, and consecutive pledges by the same firm appear to have smaller effects. One possible reason for the larger effects is that the first pledge tends to be more informative or more ambitious than subsequent pledges.

Another possibility is that our identified first pledges are less likely to contain measurement error, and therefore the estimates would be less affected by attenuation bias. Either way, this simple filter appears quite effective in identifying the most meaningful announcements.

While our results are based on green pledges identified with OpenAI's model GPT-4, many newer models have been developed since we wrote the first version of this paper. To assess the robustness of our main event study results, we use four more recent models by OpenAI, Anthropic, and Google to identify green pledges from our news articles, and also use a majority-vote label using all five models. [Internet Appendix Table B.6](#) shows that across all models, the stock market effects of green pledges are significantly positive and of roughly similar magnitude.

An important question is whether the observed stock market reactions to green pledges truly reflect the pledges themselves, or whether they are instead driven by generally positive reactions to any environmental news. To answer it, we carry out a type of placebo test, and estimate the effect of corporate environmental news that were not classified as green pledges by adding to regression (1) a separate event dummy that captures whether firm i on trading day t was covered by another, non-pledge environmental news article. The key question is whether the coefficient on the green pledge dummy is larger than the coefficient on this additional "other environmental news" event dummy. Panel A of [Table 7](#) shows the results for regressions with all green pledges in the first two columns, and with only first green pledges in the last two columns. The coefficient on the dummy for other environmental news is significantly positive in all four regressions. Apparently, any environmental news is "good news" on average and tends to increase a company's stock return, potentially due to an attention effect ([Chan 2003](#)). But the coefficient on the green pledge event dummy is two to four times larger than the coefficient on the other-news indicator. Tests for the equality of the coefficients on the two indicator variables reject the null hypothesis with p -values between 0.01 and 0.07. These estimates confirm that the stock market effect of green pledges is significantly stronger than the "placebo effect" of environmental news coverage. The particularly pronounced differences for the case of first green pledges again support the notion that these event indicators contain the most positive information for investors because they are better measures of substantial decarbonization announcements.

Because green pledges can coincide with other significant firm- or industry-specific announcements, or days with strong industry performance, it is possible that our estimates pick up these confounding factors and pledges are not actually the primary driver of the stock market response. To address the issue of confounding firm-level news, we construct "clean" green pledge events by excluding all green pledges that coincide with any other news announcements (environmental or other)

Table 7
Green pledges vs. other news

	All green pledges		First green pledges	
	(A) Other environmental news			
Green pledge	0.13*** (0.04)	0.20*** (0.04)	0.32*** (0.10)	0.33*** (0.10)
Other env. news	0.06*** (0.02)	0.10*** (0.02)	0.06*** (0.02)	0.11*** (0.02)
Number of obs.	14,788,589	17,511,046	14,788,589	17,511,046
Number of pledges	5,012	5,603	842	990
p-value	0.069	0.019	0.013	0.027
Firm controls	Y	N	Y	N
Day FE	Y	Y	Y	Y
Industry FE	Y	Y	Y	Y
	(B) Clean event window			
Green pledge	0.10** (0.05)	0.18*** (0.05)	0.22* (0.12)	0.27** (0.12)
Number of obs.	14,788,589	17,511,046	14,788,589	17,511,046
Number of pledges	2,011	2,281	407	499
Firm controls	Y	N	Y	N
Day FE	Y	Y	Y	Y
Industry FE	Y	Y	Y	Y
	(C) Industry effects			
Green pledge	0.13*** (0.03)	0.13*** (0.04)	0.29*** (0.10)	0.30*** (0.10)
Industry return		0.45*** (0.01)		0.45*** (0.01)
Number of obs.	14,788,589	14,788,589	14,788,589	14,788,589
Number of pledges	5,012	5,012	842	842
Firm controls	Y	Y	Y	Y
Day FE	N	Y	N	Y
Industry-by-day FE	Y	N	Y	N

Panel regressions for daily stock returns, controlling for potentially confounding news. Panel A includes the green pledges event dummy and a dummy for other environmental news that are not classified as a green pledge (26,821 events). The last row reports *p*-values for testing the hypothesis that the two dummy coefficients, for green pledge events and other environmental news, are equal. Panel B uses the green pledges event dummy but excludes events that had other news on the same day. Panel C uses the green pledges event dummy and additionally controls for industry-level returns or industry-by-day fixed effects. Standard errors, clustered by firm and day, are given in parentheses. **p* < .1; ***p* < .05; ****p* < .01. Sample period: January 2005 to December 2023.

on the same day.²⁸ Panel B of Table 7 reports the stock market regression results for this reduced sample of green pledges. Across specifications, the coefficients on the green pledge dummy remain positive and of broadly similar magnitude to that in Table 6, although they become slightly smaller and the standard errors larger due to the substantially reduced number of pledges relative to our full set of events.²⁹ Second, we examine whether

²⁸ That is, we consider all news articles in the Dow Jones data set, not just environmental news. Using this broadest set of news coverage, we aim to minimize the risk that other nonenvironmental announcements, including earnings announcements, could bias our results.

²⁹ We also checked whether our estimates might be affected by earnings announcements, which have long been known to drive stock prices (Beaver 1968). When we dropped all observations from our sample where green pledges coincided with earnings announcements on the same days, the estimates were essentially identical to our baseline results in Table 6. Thus, earnings announcements do not appear to be relevant for our findings.

industry-specific events or periods of strong industry performance affect our results. One specific concern is that firms might strategically time their pledges during such periods of positive industry news. To check whether the estimated stock market response is confounded by industry news, we control for industry returns in two different ways, using the 48 Fama-French industries either to include industry-by-day fixed effects or own-industry returns as an additional regressor. Panel C of Table 7 shows that neither specification materially affects the coefficient on the green pledge dummy, alleviating the concern about industry-level confounding factors.

Internet Appendix B reports the results of several other robustness tests. For two other subsets of our green pledge events—either using pledges in news articles that tag only a single firm (ISIN), or requiring at least a 30-day distance between two consecutive pledges by a firm—the estimated effects are very similar to those for the sample of all green pledges, as shown in Internet Appendix Table B.4. We also investigate whether our estimates change if we split the sample with the Paris Agreement in December 2015 since increased climate concerns since the Paris Agreement might have altered the sensitivity of the carbon premium and stock prices to green pledges. The estimates in Internet Appendix Table B.5 suggest that green pledges had positive effects on stock prices in both sample periods. While more granular sample splits cause problems with statistical power, overall, our results seem to be quite robust across different sample periods.

Our analysis so far has focused on the contemporaneous effects of green pledges. We now turn to estimates of the dynamic effects before and after the events, in order to understand (a) whether there might be leakage of news prior to the green pledge announcements, and (b) whether there are lagged effects, and potentially a partial reversal, after the announcements. We add five leads and lags of the event dummy and estimate the new panel regression

$$R_{it} = \sum_{j=-5}^5 \beta_j d_{i,t-j} + \gamma X_{it} + \alpha_s + \delta_t + \epsilon_{it}, \quad (2)$$

where $d_{i,t-j} = 1$ implies that there was a green pledge event j days before the return observation on day t .

Figure 3 shows estimates of the coefficients β_j in the top two panels, and the cumulative effects in the bottom two panels, together with 95%-confidence intervals based on clustered standard errors. The left two panels correspond to the regression using all green pledges, and the right two panels correspond to the regression using only first green pledges. Our estimates show little evidence for information leakage prior to green pledge events. For the sample with all green pledges, there is a moderate upward drift in the cumulative effects leading up to the event, but none of the

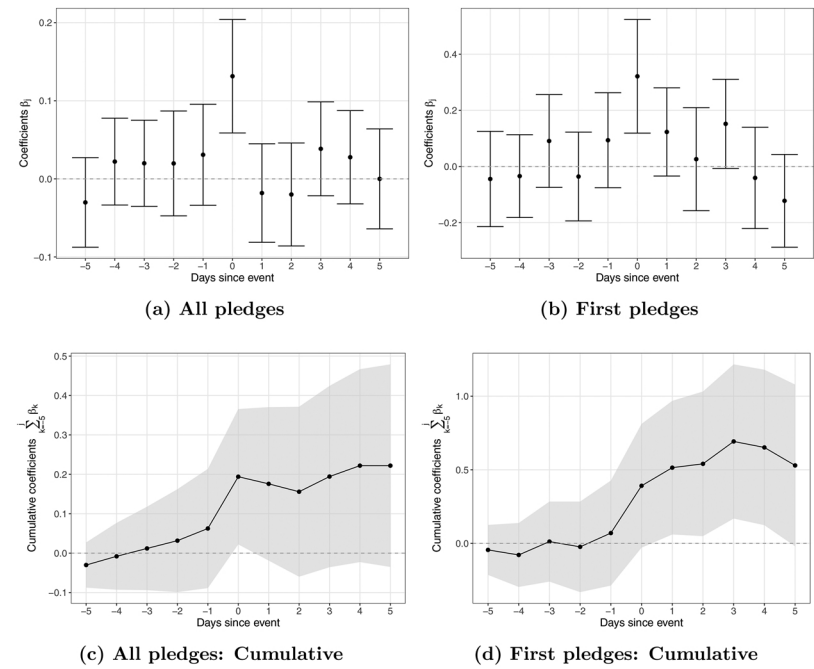


Figure 3
Stock return response around green pledges on daily stock returns
The dependent variable is the daily stock return, and the independent variable is the green pledges event dummy. The top two panels plot the coefficients β_j of regression (2) for leads and lags of the event dummy. Panel (a) shows results for the sample with all green pledges and panel (b) for only first green pledges. Controls are book-to-market, sales growth, leverage, size, and return on equity. The bottom two panels show the cumulative effects. All plots show 95% confidence intervals based on standard errors clustered by firm and day and are shown around the point estimate. Sample period: December 2005 to December 2023.

coefficients β_j for $j < 0$ are statistically significant at the 5% level. For the sample with first green pledges, there is no noticeable pre-event drift at all. The positive effect on the announcement days is not reversed in the subsequent days. Instead, the positive valuation effects continue to slightly increase over the days after the announcement; some of the coefficients for the effects several days later are positive and marginally statistically significant. Overall, the estimates in Figure 3 suggest that green pledges lead to a persistent increase in the stock market valuation, consistent with a reduction in the carbon premium and the firm’s cost of capital.³⁰

We now examine how the specificity and ambition of the announced emission reduction plans influence the financial market response to

³⁰ Figures B.3 and B.4 and in Internet Appendix show the corresponding results for four alternative specifications: (i) firm fixed effects, (ii) no controls, (iii) the sample of pledges that only tag a single firm, and (iv) the sample of pledges that requires a 30-day distance between two consecutive pledges. The results are very similar to those shown in Figure 3.

Table 8
Stock market reaction to green pledges based on different categories and ambitions

	(1)	(2)	(3)	(4)	(5)	(6)
Investment	0.15*** (0.04)	0.16*** (0.04)	0.22*** (0.04)			
Divestment	0.10 (0.24)	0.12 (0.24)	0.07 (0.24)			
Other	0.04 (0.10)	0.05 (0.10)	0.09 (0.10)			
Ambition low				0.04 (0.08)	0.05 (0.08)	0.08 (0.08)
Ambition high				0.15*** (0.04)	0.17*** (0.04)	0.23*** (0.04)
Number of pledges	5,012	5,012	5,603	5,012	5,012	5,603
R ²	0.19	0.19	0.18	0.19	0.19	0.18
Industry FE	Y	N	Y	Y	N	Y
Firm FE	N	Y	N	N	Y	N
Firm controls	Y	Y	N	Y	Y	N

Panel regression for daily stock returns, with green pledges event dummies for different categories and ambition scores as regressors. Three categories of green pledges: investments in new technologies (4,943 green pledges), divestments of pollutive assets (74 green pledges), and other pledges with no specific information on how emission reductions are achieved (587 green pledges). Two levels of ambition: low (788 green pledges) and high (4,816 green pledges). Firm controls are book-to-market, sales growth, leverage, size, and return on equity. Industry fixed effects are based on Fama-French 48 industries. All regressions include day fixed effects. Standard errors, clustered by firm and day, are given in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$. Sample period: January 2005 to December 2023.

green pledges using the categories of pledges documented in Table 2. We estimate event study regressions with three separate indicator variables that capture investment pledges, divestment pledges, and pledges without specific information on how emission reductions will be achieved. The results, reported in Table 8, show that investment pledges are followed by a significantly positive stock market reaction, suggesting that investors view such strategies as credible means of reducing emissions. Divestment pledges yield positive but imprecisely estimated and therefore statistically insignificant coefficients, due to the small sample size of only 74 divestment pledges and the resulting lack of statistical power. Notably, other, unspecific pledges are not associated with positive stock market reactions, indicating that investors recognize the limited credibility of such commitments and may view them as attempts at greenwashing. We further analyze pledges by ambition level. Pledges with high ambition scores elicit a significant and strong positive stock market reaction, while low-ambition pledges, which are likely to have only marginal effects on emissions, show no significant effect on firm valuations. Taken together, these findings provide further evidence that markets reward credible and ambitious commitments to reduce future emissions, whereas vague or low-impact pledges are discounted by investors as largely symbolic.

Finally, we turn to the issue of heterogeneity and investigate whether the stock market effects of climate commitments depend on the industry of the firm. Differential effects of climate-conscious investing play a central role

Table 9
Stock market reaction to green pledges across industries

Industry	Green Pledge		First Pledge	
	Coefficient	Standard Error	Coefficient	Standard Error
Utilities	0.087**	0.036	0.271	0.218
Energy	0.164	0.131	0.571*	0.337
Chemicals	0.472**	0.208	1.268**	0.640
Other	0.026	0.068	0.035	0.292
Manufacturing	0.027	0.123	0.640*	0.344
Nondurables	0.141	0.102	0.221	0.318
Retail	0.399**	0.195	0.126	0.252
Telecommunication	0.071	0.164	−0.549	0.680
Durables	−0.003	0.184	0.639	0.703
Business Equipment	0.285***	0.105	0.280	0.197
Health Care	−0.032	0.231	−0.030	0.466
Finance	−0.069	0.104	−0.109	0.184

Event study regressions for daily stock returns (1) estimated separately for each industry. Results are shown for the sample of all pledges as well as for only first pledges. Controls are book-to-market, sales growth, leverage, size, and return on equity. All regressions include day fixed effects. Standard errors clustered by firm and day in each model are given in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$. Sample period: January 2005 to December 2023.

for the transition to a low-carbon economy. For example, if green pledges really imply a reduced exposure to transition risks, we would expect to see an especially strong market reaction for firms in brown industries, due to the fact that they have larger transition risk exposures to begin with. If, however, mainly firms in green industries see reductions in the carbon premium and hence their cost of capital, green investing could be ineffective or even counterproductive as brown firms might respond to a relatively higher cost of capital by increasing emissions to realize short-term profits (Hartzmark and Shue 2023).

To examine industry-level heterogeneity, we estimate Equation (1) separately for each of the 12 Fama-French industries described in Table 3. We use two specifications, one with all green pledges and one with first pledges only, including standard firm controls and day fixed effects. The results are shown in Table 9, where industries are again ordered by emissions. The positive stock price response tends to be particularly pronounced in the most carbon-intensive sectors. While the statistical significance varies, in part due to substantial differences in the number of firms and pledges across industries, the estimated effect is especially strong in the utilities, energy, and chemicals sectors, the most emission-intensive industries. By contrast, in low-emission industries such as finance and health care, the coefficients are negative and statistically insignificant. The evidence aligns with the interpretation that green pledges in carbon-intensive sectors have the greatest value for investors.

The positive estimated effects of green pledges on stock prices show that investors attribute a value to decarbonization that outweighs the perceived costs and risks, and that they view the news of green pledges as a net-positive. This leads to the question whether green pledges raise valuations

by lowering risk premia and thus discount rates, or by raising expected future dividends.

The most straightforward explanation for the positive stock market effects is a discount rate channel based on changes in the carbon premium, the expected excess return on brown stocks relative to green stocks (Pastor, Stambaugh, and Taylor 2021). Indeed, the climate finance literature has generally focused on the carbon premium hypothesis to explain differences between green and brown stock returns.³¹ By announcing decarbonization commitments, firms can lower the expected path for future emissions, and reduce their carbon premium via two different effects. First, emissions are a proxy measure for a firm's exposure to climate transition risk, and green pledges would reduce this exposure and the required carbon risk premium. Second, a lower path of emissions can increase the appeal of a stock for investors that have nonpecuniary green preferences, and lower the "carbon aversion premium" (Bauer, Offner, and Rudebusch 2025). By lowering its transition risk and increasing its green investor appeal, a green pledge can make the stock more attractive to investors, lowering its discount rate and raising its price.

An alternative explanation is a cash flow channel: Green pledges could raise stock valuations because they cause investors to revise upward their expectations for future earnings and dividends. Derrien et al. (2025) show that ESG news seem to affect firm value primarily by shifting analyst earnings forecasts, as opposed to discount rate effects. We address this issue using the same empirical methodology and IBES earnings expectations data, and find no evidence that green pledges affect analyst earnings forecasts. Across all horizons, the estimated effects of green pledges on EPS forecasts are statistically insignificant. We confirm this result in several robustness checks, including restricting the sample to first pledges, treating green pledges as absorbing, using log changes, and employing mean instead of median forecasts.³² In additional analysis, we show that green pledges also do not seem to materially improve future profitability or dividend growth.³³ Overall, while we cannot rule out that factors beyond the carbon premium contribute to the observed stock market response—and the monthly frequency of the IBES data may limit the power to detect small effects on earnings expectations—our evidence suggests that green pledges are more likely to affect discount rates than expected cash flows.

³¹ See Bolton and Kacperczyk (2021, 2023) among many others. In a related study, Ardia et al. (2023) use a Campbell-Shiller decomposition of stock price effects and find that the discount rate channel is the primary channel through which climate transition risks are priced.

³² For detailed results on earnings forecasts, see Internet Appendix Table B.7.

³³ We estimate the effects of green pledges on future return on assets, return on equity, and dividend growth using the difference-in-differences methodology described in Section 4. These estimates, reported in Internet Appendix Table B.8, reveal no significant effects on ROE or dividend growth, and only small positive effects on ROA—an increase of 1.6 percentage points after 5 years.

Our stock market evidence speaks to the credibility of green pledges. The significant price effects suggest that investors do not completely discount these announcements as cheap talk and view them at least partly as credible. We have documented that the effects on stock prices are particularly strong for first green pledges, for green pledges with specific investment plans, for pledges with high levels of ambition, and for firms in brown industries. One likely reason for the stronger price effects of these types of pledges is that they cause larger reductions in expected emissions due to their stronger credibility and/or higher ambition.³⁴ On the whole, investors appear to distinguish between more and less credible green pledges, with the result that the strength of the price impact depends on the degree of credibility.

A related question is what enforcement mechanisms or incentives could prevent firms from making decarbonization commitments that are never followed by any actions and emission reductions. In other words, what incentives prevent cheap talk and make green pledges credible? Research in corporate finance going back to [Diamond \(1989\)](#) has documented the importance of reputation and disclosure quality for financing conditions and market value ([Healy and Palepu 2001](#)). In the context of climate-related and ESG disclosures, [Lyon and Maxwell \(2011\)](#) explain theoretically how reputation and possible monitoring can discipline cheap talk in climate commitments, and [Krüger \(2015\)](#) shows that negative CSR events, including failure to live up to stated goals, are punished by financial markets. On the other hand, recent work by [Jiang, Kim, and Lu \(2025\)](#) suggests limited accountability and low awareness, with failures to meet climate targets hardly covered by the media. More evidence on the financial consequences of walking back decarbonization commitments is surely needed to better understand the related incentives and consequences. Nevertheless, there are well-documented mechanisms and empirical findings indicating significant incentives for firms to follow through on their green pledges. Our finding that ambitious pledges with specific investment plans generate the strongest market reactions ([Table 8](#)) is consistent with the view that markets can distinguish credible commitments from cheap talk, providing incentives for credible corporate green pledges.

In sum, our findings show that green pledges lead to higher stock market valuations, likely because they reduce firms' carbon premia and cost of capital. This market response creates an incentive for firms to commit to decarbonization and may further facilitate the financing of green investments. This evidence points to a positive role of climate-related

³⁴ In addition, stocks in brown industries may see larger price effects because they are generally more exposed to transition risk and include a larger carbon premium.

investing in supporting firm-level decarbonization and the transition to a low-carbon economy.

4. Future Emissions

The significantly positive stock market effects of corporate green pledges suggest that investors tend to view these pledges as partly credible signals of future emission reductions. But are the announced commitments really followed by reductions in firm-level emissions? In the words of [Bingler et al. \(2022\)](#), do firms “walk the climate talk” and follow up their climate commitments with measurable actions? This question is particularly pressing given increasing concerns about greenwashing and cheap talk in climate-related announcements and disclosures ([Nemes et al. 2022](#); [Bingler et al. 2022, 2024](#); [Dzieliński et al. 2023](#)). Companies may falsely represent themselves as environmentally friendly by manipulating environmental metrics or rebranding products and marketing strategies touting their clean energy or pollution reduction efforts. And their green pledges might just be cheap talk, without meaningful subsequent reductions in emissions.

To address this question, we estimate the response of future firm-level emissions to announcements of corporate green pledges. We use difference-in-differences (DiD) estimation, which compares changes in emissions before and after the issuance of a green pledge to the changes in emissions for firms that have not made such pledges. Since firms decide strategically whether and when to announce a green pledge, instead of pledges being an exogenous treatment, this estimation naturally does not identify the causal effects of the announcements themselves. Instead, the estimates capture the extent to which green pledges predict changes in future emissions, which is precisely the type of new information that stock market investors would find relevant. Put differently, while pledges do not cause emission reductions, they reveal credible private information about future emission trajectories that investors value and incorporate into prices.³⁵

Two aspects of our empirical setting complicate the accurate estimation of the “treatment effects”—or more accurately, of the information content—of green pledges for future emissions. Sticking with the common

³⁵ It is hard to imagine a setting in which a corporate announcement could be exogenous, as it is always the endogenous result of strategic planning and, in most cases, shareholder value maximization. For example, a firm may well have had long-standing plans for decarbonization with the goal of increasing the firm's value, and the public release of these plans is merely the last step before their implementation. Through the lens of DiD estimation, the parallel trends assumption is naturally violated in our setting since firms with green pledges are likely on a different emission trajectory from other firms. We are estimating the differences in emission trends between the two types of firms, which is not a causal effect of the announcement but nevertheless the relevant object of interest for predicting emissions. Examples of other financial research using DiD for prediction and other purposes besides causal inference are surveyed by [Gow, Larcker, and Reiss \(2016\)](#).

terminology in the DiD literature, the “treatments” are, of course, staggered, as different firms announce their green pledges at different times. While variation in treatment timing is by itself not necessarily problematic, the combination with heterogeneous treatment effects can render standard event study estimates unreliable. Since decarbonization commitments may differ widely in terms of their ambition, specificity, and timelines, but are all captured by a simple binary indicator, treatment effects should indeed be expected to be heterogeneous in our context. With staggered treatments and heterogeneous treatment effects, standard two-way fixed effects (TWFE) estimates may yield inconsistent estimates of the average treatment effect (Baker, Larcker, and Wang 2022; Roth et al. 2023). The problem is the “forbidden” comparisons between newly treated units and previously treated units, as the latter may still be experiencing delayed treatment effects. Various approaches have been proposed to estimate average treatment effects in a way that addresses the limitations of TWFE (Callaway and Sant’Anna 2021; Sun and Abraham 2021; Dube et al. 2025).

We use the local projections difference-in-differences (LP-DiD) approach by Dube et al. (2025) for our baseline estimates of the dynamic response of emissions to green pledges. This simple and reliable method accounts for heterogeneous treatment effects by including only “clean controls,” that is, firms that have not yet been treated themselves at the time of each treatment under consideration. In a firm-by-year panel, we estimate the LP-DiD specification

$$y_{i,t+h} - y_{i,t-1} = \beta^h \Delta D_{it} + \sum_{p=1}^P \gamma_p^h \Delta y_{i,t-p} + \sum_{m=1}^M \sum_{p=0}^P \gamma_{mp} \Delta x_{m,i,t-p} + \delta_t^h + \epsilon_{it}^h, \quad (3)$$

where the indicator D_{it} captures whether firm i has issued a green pledge in year t or before. Since the indicator D_{it} equals one in the year of its first pledge and all subsequent years, the treatment is “absorbing”—the most common assumption in difference-in-differences estimation—and we estimate the effects of each company’s first green pledge. The sample is restricted to the observations that are either newly treated ($\Delta D_{it} = 1$) or clean controls ($D_{i,t+h} = 0$). The variable y_{it} denotes either log carbon emissions or emission intensity, in both cases based on the sum of Scope 1 and Scope 2 emissions. The regression estimates the effects of green pledges on log emissions or emission intensity h years after the pledge. Our LP-DiD regression (3) also includes year fixed effects, δ_t^h , lagged emissions, and current and lagged values of M different controls. Lagged values of $y_{i,t}$ are included to account for the pattern documented above that brown firms are more likely to commit to decarbonization. We control for firm characteristics that could be correlated with the treatment, including size, book-to-market ratio, leverage, profitability, revenue growth, and log

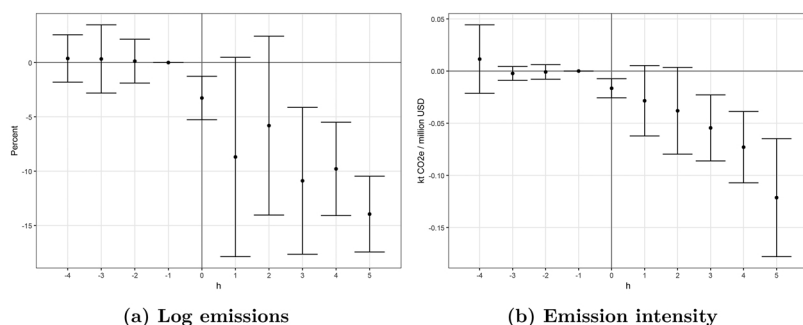


Figure 4

Impact of green pledges on carbon emissions

Local projection difference-in-differences regression estimates with the log-level of emissions (left panel) or emission intensity (right panel) as the dependent variable and green pledges as the independent variable. Controls include lagged values of size, book-to-market, leverage, profitability, revenue growth, and log PP&E, as well as lagged emissions (log levels or intensities). Error bars correspond to 95% confidence intervals based on [Driscoll and Kraay \(1998\)](#) standard errors.

PP&E. By the nature of the LP-DiD estimation, the controls are included as first differences, and we set $P = 2$ for the number of lags.

[Figure 4](#) plots the estimated effects on log emissions (left panel) and emission intensities (right panel) from 4 years before the pledge to 5 years after. Firms that make a decarbonization commitment significantly reduce their carbon emissions after the pledge, compared to firms without such commitments. Green pledges predict 9% lower emissions after 1 year and 14% lower emissions after 5 years, compared to those without a pledge. Prior to the announcement, there is no significant difference in emissions, which alleviates concerns that firms issuing a green pledge are simply confirming an existing downward trajectory for emissions. For changes in emission intensity, we find broadly similar results, as shown in the right panel of [Figure 4](#). Following a green pledge, emission intensity falls significantly, with a decrease of 0.03 (kilotons of CO₂ equivalents per million U.S. dollars revenue) in the first year and 0.12 after 5 years. Prior to the pledges, changes in emission intensities are not significantly different between firms with and without a pledge.

A concern about firm-level emissions data is that many firms do not report emissions, and vendor estimates can be noisy and unreliable (e.g. [Aswani, Raghunandan, and Rajgopal 2023](#)). To address this concern, we estimate local projections using only firms with reported emissions, excluding observations with Trucost estimates. The estimates, shown in [Internet Appendix Figure B.5](#), show that the results remain qualitatively unchanged, and, in fact, become slightly stronger in this subsample, alleviating concerns about an undue influence of estimated emissions.

In [Internet Appendix B](#), we also use LP-DiD estimation for other dependent variables to try to understand the predictive power of green pledges for future investment and profitability (see [Internet Appendix Table B.8](#)).

As discussed in [Section 3](#), these results are helpful for interpreting the stock market results and are consistent with the explanation that green pledges affect stock prices primarily via changes in the carbon premium.

The LP-DiD approach is one of several methods designed to estimate the average treatment effect (ATE) in the presence of staggered treatment and heterogeneous treatment effects. [Sun and Abraham \(2021\)](#) modify TWFE estimation by including only clean controls and incorporating a cohort-specific dummy variable to capture heterogeneous treatment effects across cohorts, and then estimate the ATE using the appropriate weighted average of cohort-specific effects. These and alternative approaches proposed in the literature ultimately aim to estimate the same object of interest, the ATE, which in our setting is the difference between a firm's emissions after making a green pledge and the emissions it would be expected to produce without the pledge. This common objective allows for a meaningful comparison of the estimated effects across different methods. [Table 10](#) shows such a comparison, including results for LP-DiD, conventional TWFE estimation, and the [Sun and Abraham \(2021\)](#) (S&A) method. The top panel reports the estimated effects on log emissions, and the bottom panel shows the effects on emission intensities. We obtain estimates for two different time horizons—1 year and 5 years after the pledge—and both for regressions with and without controls. All the different estimation results consistently show a decrease in emissions and intensities following a green pledge. While point estimates differ somewhat across methods, the estimated reduction is generally substantive and in the majority of the cases statistically significant. [Table 10](#) also shows that controlling for pre-treatment firm characteristics only slightly reduces the estimated magnitudes. Overall, these estimates demonstrate the robustness of our main results across different estimation methods and specifications.

A related question is which firms actually reduce their emissions, and whether these are the same firms that experience the strongest stock market reactions. To address it, we estimate the LP-DiD regressions separately for each industry. [Table 11](#) shows the estimated coefficients for 5-year changes in emission levels (in percent) and emission intensity across the 12 Fama-French industries considered above. Firms in the two sectors with the highest carbon emissions—utilities and energy—exhibit the most substantial reductions in both emission levels and emission intensities following the green pledges, consistent with the positive stock market effects documented in [Table 9](#). Besides these two particularly brown industries, the pattern is less clear and there is a fair amount of heterogeneity in the industry-level estimates that is not evidently related to industry brownness.³⁶ Nevertheless, there is some evidence that in the

³⁶ The retail sector is somewhat puzzling, as it shows both a significantly positive stock market reaction to green pledges and a significant increase in emission intensity. This pattern likely reflects sector-specific

Table 10
Average treatment effect of green pledges on carbon emissions

	LP-DiD		TWFE		S&A	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>(A) Log emissions</i>						
1 year	−8.69*** (3.32)	−10.23*** (3.68)	−3.36 (3.47)	−6.86 (5.80)	−3.51* (2.03)	−8.57*** (2.46)
5 years	−13.96** (6.52)	−9.84 (6.57)	−5.15 (4.17)	−6.07 (6.56)	−6.25** (2.77)	−7.53 (5.62)
Observations	7,987	14,165	12,609	21,511	12,609	21,511
<i>(B) Emission intensity</i>						
1 year	−0.027* (0.014)	−0.030 (0.034)	−0.011 (0.032)	−0.054 (0.039)	−0.015 (0.018)	−0.041** (0.020)
5 years	−0.121*** (0.032)	−0.104*** (0.033)	−0.054 (0.043)	−0.100* (0.052)	−0.080 (0.050)	−0.091** (0.043)
Observations	7,986	14,160	12,608	21,502	12,608	21,502
Controls	Y	N	Y	N	Y	N
Year FE	Y	Y	Y	Y	Y	Y
Firm FE	N	N	Y	Y	Y	Y

Alternative estimates of the average “treatment effect” of green pledges on 1- and 5-year-ahead log level of emissions (panel A) or emission intensity (panel B). Columns 1 and 2 show LP-DiD estimates following [Dube et al. \(2025\)](#), and the first column corresponds to the estimates in [Figure 4](#) for $h = 1$ and $h = 5$. Columns 3 and 4 report standard two-way fixed effects estimates. Columns 5 and 6 show estimates using the [Sun and Abraham \(2021\)](#) approach. Estimates for log emissions are in the top panel, and estimates for emission intensities are in the bottom panel. Standard errors are Driscoll-Kraay for LP-DiD, and clustered (by year and firm) for TWFE and S&A. * $p < .1$; ** $p < .05$; *** $p < .01$. The number of observations corresponds to the 1 year horizon. Sample period: January 2005 to December 2023.

brownest industries, which are also those with the strongest stock market reactions, as documented in [Section 3](#), green pledges are particularly useful and credible signals for future emission reductions.

There are various ways for firms to reduce their emissions. As discussed above in [Section 1](#), there are two fundamentally different strategic directions: Companies can invest in green technologies and in this way lower emissions in their production processes. Alternatively, they might divest from certain high-emission business lines, potentially selling these to other companies. In [Section 1](#) we showed that in the vast majority of green pledges, the text of the announcement indicates plans for additional investment. Consistent with this pattern, additional LP-DiD estimates show that after announcing green pledges, firms tend to significantly increase their investment.³⁷

Overall, corporate green pledges appear to contain relevant new information and they lower the expected trajectory for future firm-level

factors: Retail firms have relatively low direct emissions, making emission intensity measures sensitive to small changes in emissions or revenues. In addition, many retail pledges focus on supply chains and logistics (Scope 3 emissions), while our analysis relies on Scope 1 and 2 emissions. Finally, transition investments in logistics or store infrastructure may temporarily increase energy use before reductions materialize.

³⁷ [Internet Appendix Table B.8](#) shows LP-DiD estimates of the effects of green pledges on capital expenditure at 1- and 5-year horizons, which are significantly positive.

Table 11
Emission changes across industries

Industry	Carbon Emission		Emission Intensity	
	Coefficient	Standard Error	Coefficient	Standard Error
Utilities	−43.676***	10.057	−0.031***	0.007
Energy	−51.754***	6.833	−0.010	0.017
Chemicals	12.557	12.077	0.002***	0.001
Other	14.811	13.518	0.004*	0.002
Manufacturing	2.794	3.412	−0.010***	0.003
Nondurables	−2.244	5.716	0.008	0.012
Retail	5.171	6.382	0.229**	0.105
Telecommunication	−38.065	45.098	−0.871***	0.271
Durables	−41.165***	14.774	0.002	0.002
Business Equipment	−29.006***	4.954	−0.119***	0.023
Health Care	−14.851*	8.060	−0.010***	0.003
Finance	3.062	4.806	−0.015	0.070

Local projection difference-in-differences regressions for 5-year forward log level of emissions or emission intensity, estimated separately for each of the 12 Fama–French industries. Controls include lagged values of size, book-to-market, leverage, profitability, revenue growth, and log PP&E, as well as lagged emissions (log-levels or intensities). All regressions include year fixed effects. Standard errors are based on [Driscoll and Kraay \(1998\)](#). * $p < .1$; ** $p < .05$; *** $p < .01$. Sample period: January 2005 to December 2023.

emissions. This finding helps rationalize the significant positive stock market response to green pledges documented in [Section 3](#). The estimated decline in emissions suggests that investors have good reasons to view corporate green pledges as credible, thereby mitigating concerns about greenwashing and cheap talk in climate commitments.

5. Conclusion

The transition to a net-zero economy poses a major challenge for corporations, including regulations that make CO₂ emissions costly and could render high-carbon business models obsolete. By making commitments to reduce carbon emissions, companies can lower their exposure to such transition risks and make their stocks more attractive to investors. However, decarbonization strategies can be costly and the outcomes are risky. Furthermore, announcements of planned decarbonization may be regarded as cheap talk instead of credible signals about future emissions. Whether such announcements have any effects on stock prices, and whether they are in fact followed by changes in emissions, are important empirical questions in climate finance.

Using a new database of corporate climate commitments derived from news articles, this paper studied the stock market effects of these commitments. Our event study results showed that decarbonization commitments lead to a significant increase in stock prices, indicating that the positive effects on company valuations outweigh the costs and risks from decarbonization. Our evidence is consistent with the carbon premium hypothesis of [Pastor, Stambaugh, and Taylor \(2021\)](#) and suggests that companies are able to reduce their transition risk exposure, appeal

to green investors, and ultimately lower their carbon premium by issuing green pledges.

Corporate green pledges do not appear to be cheap talk, but are instead followed by significant reductions in firm-level emissions and emission intensities. This result rationalizes the positive stock market reaction because investors have good reasons to view green pledges as credible. It also suggests that the financial incentive for climate commitments, in the form of higher stock valuations and lower cost of capital, appears to be justified, as the corporate announcements are followed by corporate climate actions. Given that these voluntary commitments are also more prevalent for large and brown firms, they have the potential to meaningfully contribute to the green transition of the U.S. economy.

Our work opens up several avenues for future research. First and foremost, our method for identifying and classifying green pledges could be improved, for example, by using more sophisticated prompting strategies or new types of models, including those with reasoning abilities. Future work could also develop additional criteria for evaluating the ambition, urgency, specificity, and credibility of the announced commitments. Because of the global scale of the issue, another natural extension of our work is to incorporate data from other countries, in particular those of the European Union, given their ambitious climate goals and the availability of high-quality data on firm emissions. A central open question is whether and how firms are held accountable for their green pledges and climate targets. More direct evidence on the financial consequences of failing to meet emission targets is needed to better understand the incentives for firms to follow through on their climate commitments. More broadly, future research should address the question of whether, on aggregate, corporate climate commitments are sufficient to decarbonize the economy or quantify the shortfall between corporate commitments and national net-zero goals. Recent results for the U.S. corporate sector from [Pastor, Stambaugh, and Taylor \(2024\)](#) suggest that there is a significant shortfall relative to the goals set forth in the Paris Agreement.

Code Availability: The replication code is available in the Harvard Dataverse at <https://doi.org/10.7910/DVN/88JIH6>.

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